

AI Engineering Lab

Answer Key — Quiz Form B

The correct option of each item is set in bold. Each item also shows its blueprint cell and the misconception its distractors target.

Internal document — do not hand out to students.

Part C Knowledge questions

20 questions, each with exactly one best answer. Tick one box per question. If you do not know, please tick “I do not know” rather than guessing — an honest blank is more useful to us than a lucky guess, and none of this affects your grade. Plan for about 20 minutes.

1. Which statement about the LLM inside an agentic system is correct?
- (a) It accumulates knowledge about the current task across the calls of one run.
 - (b) It keeps the tool results it has already seen, so they need to be sent only once.
 - (c) **It has no memory of previous calls, so the harness has to resend everything it needs.**
 - (d) It remembers the system prompt after the first call and does not require it again.
- ☐ I do not know.

[topic: T1 — LLM role & statelessness | level: remember | difficulty: easy | blueprint cell: B01]

Solution: All three distractors are forms of the same error — assuming server-side state — and each is a bug students actually write when assembling requests.

2. A team observes that their agent follows instructions reliably in short runs but starts ignoring them in long ones, although the context window is never exceeded. What is the most plausible explanation?
- (a) The system prompt is silently truncated once the conversation passes a certain number of messages.
 - (b) Instructions expire after a fixed number of turns and have to be repeated.
 - (c) The model switches to a smaller variant automatically to keep response times constant.
 - (d) **The model’s effective attention to any single instruction falls as the window fills up with lower-signal material.**
- ☐ I do not know.

[topic: T2 — Context engineering | level: understand | difficulty: medium | blueprint cell: B02]

Solution: Same cell as A02: degradation begins well before the hard limit. The distractors invent mechanisms that would produce an abrupt rather than a gradual failure.

3. Your harness summarises the oldest third of the conversation whenever the context exceeds a threshold. What are you accepting in exchange?
- (a) The agent can no longer call any tool that was used in the summarised part of the run.
 - (b) Detail from the summarised turns is no longer available in full, so some fidelity is traded for sustained coherence and lower cost.**
 - (c) The summary has to be produced by a different model from the one running the loop.
 - (d) The loop's termination conditions stop applying, because the iteration count is reset.
- ☐ I do not know.

[topic: T2 — Context engineering | level: apply-analyse | difficulty: hard | blueprint cell: B03]

Solution: Same cell as A03, approached from the cost side rather than the choice side: compaction is a deliberate loss of fidelity, and knowing what you gave up is the point.

4. Your agent occasionally gets stuck calling the same failing tool over and over until you kill the process. Which property of the loop is missing?
- (a) A retry policy that repeats each failing tool call until it succeeds.
 - (b) A larger context window, so that the repeated observations still fit.
 - (c) A sandbox, so that repeated tool calls cannot damage the host system.
 - (d) A bound on execution — for example a maximum number of iterations — that forces the machine into a terminal state.**
- ☐ I do not know.

[topic: T3 — Loop and control flow | level: understand | difficulty: medium | blueprint cell: B04]

Solution: Same cell as A04. (a) is the opposite of what is needed and (c) attributes termination to the sandbox.

5. Some harnesses build planning instructions into the system prompt rather than waiting for a user to ask for a plan. What follows directly from that design?
- (a) Every run spends its first reasoning effort on producing a plan, whether or not the task needs one.**
 - (b) The user can no longer influence how the agent approaches the task.
 - (c) The agent's tool calls are validated against the plan by the tool registry.
 - (d) The plan replaces the conversation history as the agent's context.
- ☐ I do not know.

[topic: T3 — Loop and control flow | level: apply-analyse | difficulty: hard | blueprint cell: B05]

Solution: Same cell as A05: up-front planning is a trade, not a free improvement — it costs the first turn of every run, including trivial ones.

6. Which of these would be implemented as a tool rather than as some other component of the harness?
- (a) A bundle of instructions that teaches the agent how to fill in a particular PDF form.
 - (b) A callback that runs before every action to check whether it is permitted.
 - (c) A function that lists the files in a directory and returns the listing to the model.**
 - (d) A policy that decides whether an action requires the user's confirmation.

☐ I do not know.

[topic: T4 — Tools and tool design | level: remember | difficulty: easy | blueprint cell: B06]

Solution: Same cell as A06, asked as a classification instead of a definition: (a) is a skill, (b) a hook, (d) permission management.

7. Your harness catches every exception raised inside a tool and returns the literal string `error` to the model. Which consequence follows most directly?
- (a) The harness can no longer log the original exception, because it has been replaced.
 - (b) The model has nothing to work with in order to choose a corrective action, so it cannot recover from failures it could otherwise handle.**
 - (c) The tool's schema becomes invalid, because its declared return type is not met.
 - (d) The agent treats every failure as a final answer and ends the run.

☐ I do not know.

[topic: T4 — Tools and tool design | level: apply-analyse | difficulty: hard | blueprint cell: B07]

Solution: Same cell as A07 with the ambiguity introduced deliberately rather than accidentally: errors must reach the model in a form it can act on. Logging (a) is unaffected.

8. In the architecture of a harness, what role does an MCP server play?
- (a) It hosts the language model that the agent sends its requests to.
 - (b) It stores the agent's conversation history so that it survives a restart.
 - (c) It provides tools that the agent can discover and call over a standardised interface.**
 - (d) It enforces which of the agent's actions require the user's approval.

☐ I do not know.

[topic: T5 — MCP | level: remember | difficulty: easy | blueprint cell: B08]

Solution: Same cell as A08: (a) is the "MCP serves the model" confusion, (b) long-term memory, (d) permission management.

9. A team adds a new tool to their MCP server. Their agent picks it up on the next run with no code change and no configuration change. Which property of their integration does that demonstrate?

- (a) **Tools are discovered at runtime, so the agent's action space follows the server's.**
 - (b) The tool schemas are compiled into the harness at build time.
 - (c) The model was fine-tuned on the server's tool catalogue.
 - (d) The tool registry caches every tool it has ever seen and replays it.
- ☐ I do not know.

[topic: T5 — MCP | level: apply-analyse | difficulty: medium | blueprint cell: B09]

Solution: Same cell as A09, reasoning from an observation back to the mechanism. (b) would produce the opposite behaviour.

10. A harness holds a library of 60 skills, but a given run typically loads two or three of them. What mechanism makes that possible?

- (a) **The skills are disclosed just in time: only an identifier and a short description are visible until one is actually needed.**
 - (b) The skills are compressed and transparently decompressed by the context manager.
 - (c) The agent requests skills from the model provider's servers rather than from the harness.
 - (d) The unused skills are held in long-term memory and summarised into the context.
- ☐ I do not know.

[topic: T6 — Skills | level: understand | difficulty: medium | blueprint cell: B10]

Solution: Same cell as A10: the mechanism is selective disclosure, not compression — (b) is the common wrong intuition that the problem is size rather than signal.

11. Which requirement can be met *only* by a long-term memory component, and not by context management alone?

- (a) Older turns of the current conversation are summarised in order to save tokens.
 - (b) **Information stored during one session is still available after the application has been restarted.**
 - (c) The tool schemas stay visible to the model throughout the run.
 - (d) The agent's reasoning from the previous iteration informs the next one.
- ☐ I do not know.

[topic: T7 — Long-term memory | level: remember | difficulty: easy | blueprint cell: B11]

Solution: Same cell as A11: persistence beyond the process is the defining property. (a), (c) and (d) are all achievable within a single run's context.

12. At which points would a hook typically be triggered?

- (a) Only once, when the agent process starts up.
- (b) Whenever the user types a new message, and at no other time.
- (c) Whenever the context window is about to overflow, and at no other time.
- (d) Before a tool call and after an LLM response.**

☐ I do not know.

[topic: T8 — Hooks | level: remember | difficulty: easy | blueprint cell: B12]

Solution: Same cell as A12: hooks sit at the transitions of the loop, which is what makes them the natural place for enforcement and instrumentation.

13. Your team has to record a metric for every tool call, every model call and every permission decision, without scattering counter increments throughout the codebase. Which mechanism gives you that?

- (a) A tool that the agent calls whenever it wants something to be counted.
- (b) A wrapper around the metrics endpoint that infers the events from HTTP traffic.
- (c) Lifecycle hooks at those points, with the instrumentation attached to the hooks.**
- (d) A sub-agent whose task is to watch the main agent and report what it did.

☐ I do not know.

[topic: T8 — Hooks | level: apply-analyse | difficulty: medium | blueprint cell: B13]

Solution: Same cell as A13, using the observability side of the same seam. (a) makes measurement depend on the model choosing to measure.

14. In a model with a read-only tier, a sandbox-edit tier and a full-access tier, which action is characteristic of the sandbox-edit tier?

- (a) Writing a file inside the container's workspace.**
- (b) Sending an email on the user's behalf.
- (c) Searching the repository for a string.
- (d) Deleting a branch on the remote repository.

☐ I do not know.

[topic: T9 — Permission management | level: understand | difficulty: easy | blueprint cell: B14]

Solution: Same cell as A14, classifying into the middle tier instead of the top one: (c) is read-only, while (b) and (d) reach outside the sandbox.

15. Your permission system is configured so that every single tool call requires the user's confirmation. Which criticism of that configuration is valid?

- (a) It makes the permission decisions impossible to record in the observability data.
- (b) It gives up the benefit of the tiers: routine, low-risk observation interrupts the user as often as genuinely consequential action.**
- (c) It means the sandbox is no longer needed, since the user approves everything.
- (d) It prevents the agent from using tools discovered from an MCP server.

☐ I do not know.

[topic: T9 — Permission management | level: understand | difficulty: medium | blueprint cell: B15]

Solution: Same cell as A15 approached from over-restriction: tiers exist to spend human attention where it changes the outcome. Confirmation fatigue makes approvals reflexive and defeats the gate.

16. A team argues that because their agent runs inside a container, they do not need permission management. Why is that argument wrong?

- (a) A sandbox bounds what the agent *can* reach; permission management decides which of the reachable actions it *may* take, including ones whose effects leave the sandbox.**
- (b) Containers isolate only the network and not the file system, so file access still needs a policy.
- (c) Permission management is what starts and stops the container, so it cannot be removed.
- (d) A container's isolation expires after a fixed time, after which a policy has to take over.

☐ I do not know.

[topic: T10 — Sandboxing | level: understand | difficulty: medium | blueprint cell: B16]

Solution: Same cell as A16: can versus may. A containerised agent with a network connection and an API token can still act irreversibly on the outside world.

17. Why does giving a sub-agent its own context window help the system scale?

- (a) The two agents can call the LLM simultaneously, halving the run's latency.
- (b) The sub-agent's context does not count towards the provider's token limits.
- (c) The main agent can then run on a smaller model, since the reasoning happens in the sub-agent.
- (d) The coordinating agent's context stays focused, because only the summary of the sub-agent's work is added to it.**

☐ I do not know.

[topic: T11 — Sub-agents and A2A | level: understand | difficulty: medium | blueprint cell: B17]

Solution: Same cell as A17: the benefit is context economy, not parallelism (a) or a billing loophole (b).

18. A dashboard for an agent harness shows only the number of completed runs per hour. What is the most important thing it is missing?

- (a) The version number of the harness that produced them.
- (b) The number of lines of Python in the tool registry.
- (c) Whether those runs succeeded or failed.**
- (d) The physical location of the server that ran them.

☐ I do not know.

[topic: T12 — Observability and telemetry | level: understand | difficulty: easy | blueprint cell: B18]

Solution: Same cell as A18: throughput without an outcome dimension cannot distinguish a healthy system from one that is failing fast and often.

19. Why is it not enough to expose only aggregate metrics for an agentic system?

- (a) Aggregate metrics cannot be scraped by a Prometheus-compatible server.
- (b) Aggregate metrics are only valid if the agent is deterministic, which it is not.
- (c) Aggregate metrics are computed by the model itself and are therefore unreliable.
- (d) They show that something is wrong but not why; reconstructing what happened in one run requires per-run traces or structured logs.**

☐ I do not know.

[topic: T12 — Observability and telemetry | level: apply-analyse | difficulty: medium | blueprint cell: B19]

Solution: Same cell as A19, stated as a principle rather than an incident. This is why the lab asks for both a dashboard and inspectable per-run records.

20. Why should the assertions of an end-to-end test for an agent target the resulting state rather than the agent's wording?

- (a) Because the agent's messages are not available to the test harness.
- (b) Because the same correct outcome can be reached with different phrasings and different sequences of calls.**
- (c) Because state assertions are the only kind a test framework is able to express.
- (d) Because the wording of the agent's messages is fixed by the model provider.

☐ I do not know.

[topic: T13 — Quality assurance | level: understand | difficulty: medium | blueprint cell: B20]

Solution: Same cell as A20 as a principle: asserting on wording produces flaky tests, because the model's phrasing varies while the outcome does not.